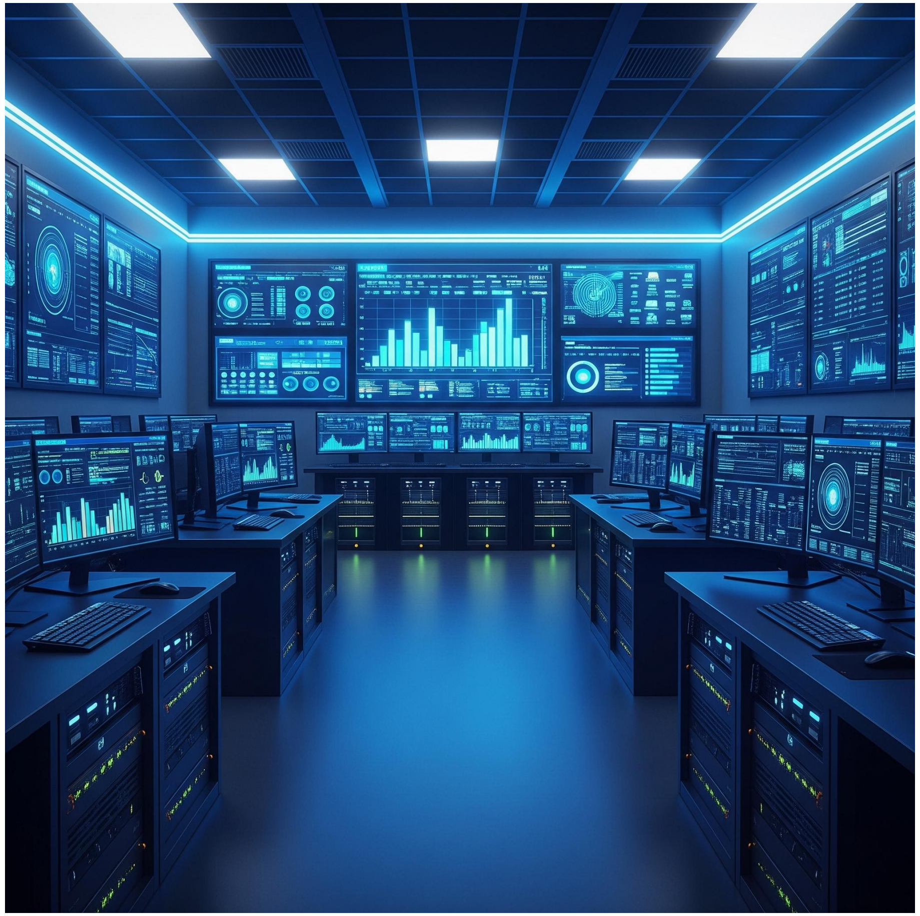
Modern SOC Implementation with Open-Source Tools

Leveraging SANS, NIST, and IEEE guidance for a next-gen SOC



Introduction & Agenda

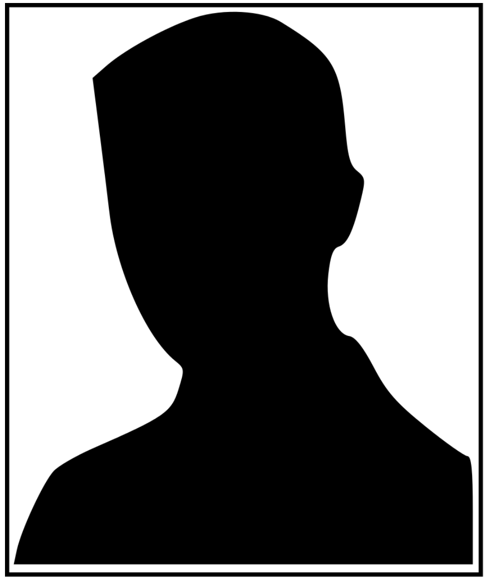
- What is a SOC?
- Core SOC Functions
- Data Sources & SIEM Role
- Legacy Challenges vs. Modern Benefits
- Open-Source Tools Stack
- Integration Case Study
- Metrics & Playbooks
- Key Takeaways

Whoami !

Aditya Firman “Ghxyss”
ilmutukang.notion.site
hobinya nukang sama makan !

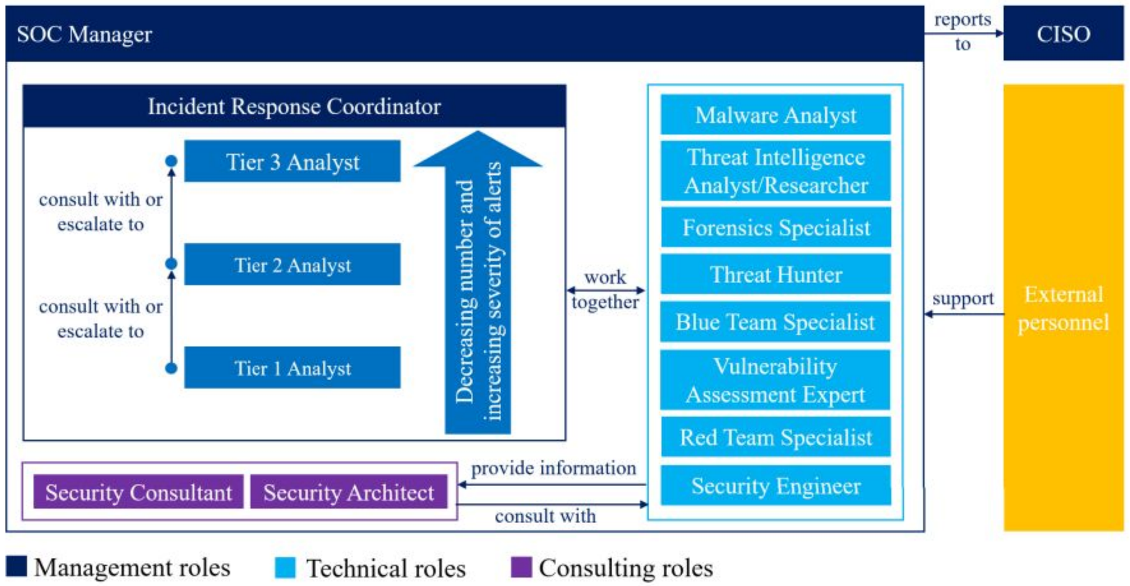
https://www.linkedin.com/in/adityyaaa/

aditya.firman@ldn.id



What is a Security Operations Center (SOC)?

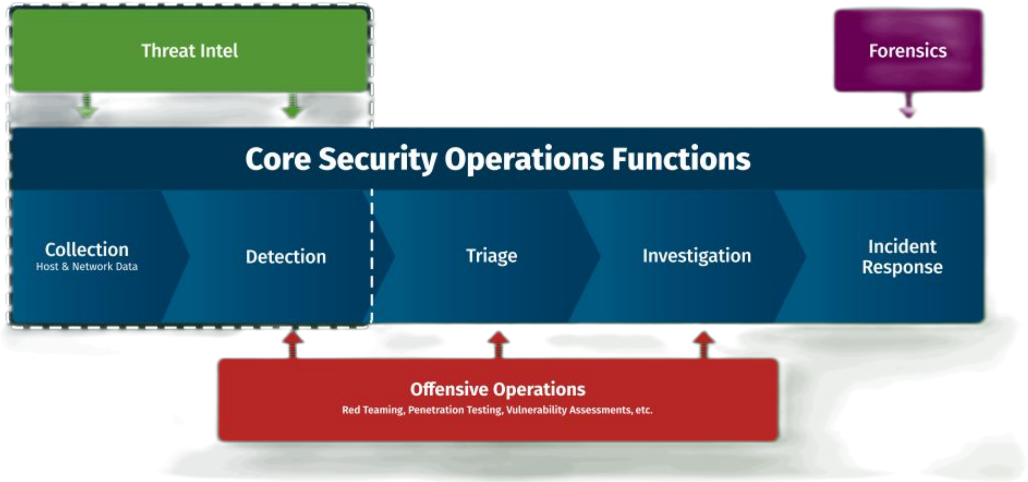
- 24/7 centralized cyber threat team
- Protects systems, data, people
- Combines people, process, tech
- Incident collaboration hub



Security Operations Center: A Systematic Study and Open Challenges Figure 3

Core SOC Functions (1)

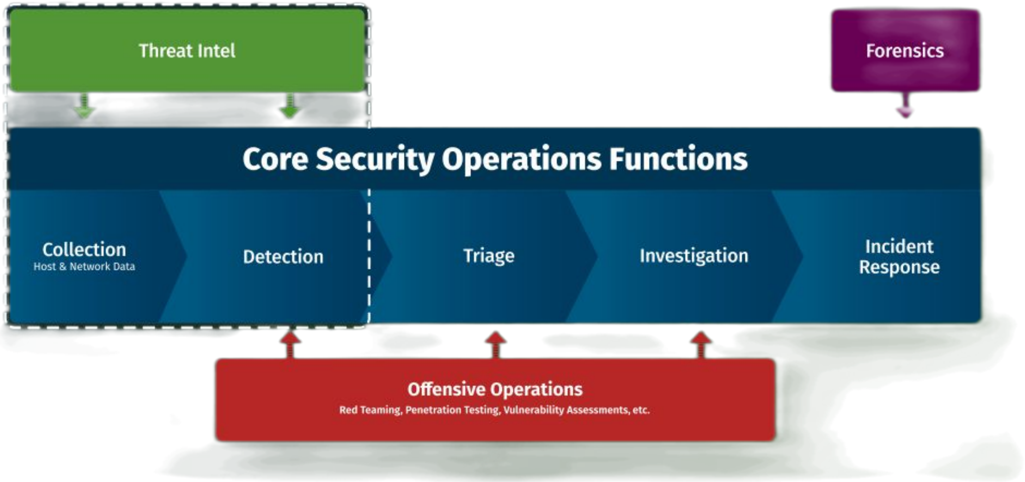
- **Collection**: Logs from endpoint, network, cloud
- **Detection**: Alerting via analytics, threat intel
- **Triage**: Prioritize alerts by severity and risk



Guide to Security Operation

Core SOC Functions (2)

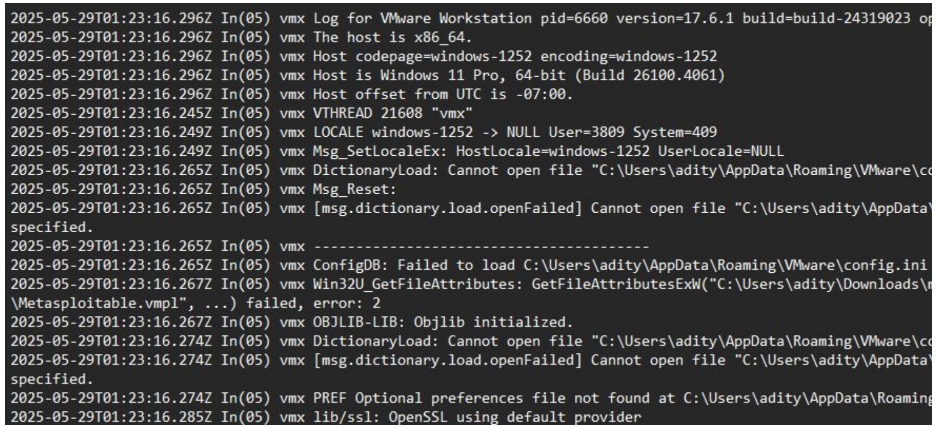
- **Investigation**: Validate and scope incidents
- **Response**: Containment, recovery
- Feedback loop for detection improvements



[Guide to Security Operation]

Key Data Sources

- Network logs, NetFlow (Suricata/Mirror Port)
- Endpoint logs (OSSEC/Wazuh)
- Cloud, virtualization
- Security tools, threat intel feeds (TIA)



```
2025-05-29T01:23:16.296Z In(05) vmx Log for VMware Workstation pid=6660 version=17.6.1 build=build-24319023 op
2025-05-29T01:23:16.296Z In(05) vmx The host is x86_64.
2025-05-29T01:23:16.296Z In(05) vmx Host codepage=windows-1252 encoding=windows-1252
2025-05-29T01:23:16.296Z In(05) vmx Host is Windows 11 Pro, 64-bit (Build 26100.4061)
2025-05-29T01:23:16.296Z In(05) vmx Host offset from UTC is -07:00.
2025-05-29T01:23:16.245Z In(05) vmx VTHREAD 21608 "vmx"
2025-05-29T01:23:16.249Z In(05) vmx LOCALE windows-1252 -> NULL User=3809 System=409
2025-05-29T01:23:16.249Z In(05) vmx Msg_SetLocaleEx: HostLocale=windows-1252 UserLocale=NULL
2025-05-29T01:23:16.265Z In(05) vmx DictionaryLoad: Cannot open file "C:\Users\adity\AppData\Roaming\VMware\c
2025-05-29T01:23:16.265Z In(05) vmx Msg_Reset:
2025-05-29T01:23:16.265Z In(05) vmx [msg.dictionary.load.openFailed] Cannot open file "C:\Users\adity\AppData
specified.
2025-05-29T01:23:16.265Z In(05) vmx ----------------------------------------
2025-05-29T01:23:16.265Z In(05) vmx ConfigDB: Failed to load C:\Users\adity\AppData\Roaming\VMware\config.ini
2025-05-29T01:23:16.267Z In(05) vmx Win32U_GetFileAttributes: GetFileAttributesExW("C:\Users\adity\Downloads\
\Metasploitable.vmpl", ...) failed, error: 2
2025-05-29T01:23:16.267Z In(05) vmx OBJLIB-LIB: Objlib initialized.
2025-05-29T01:23:16.274Z In(05) vmx DictionaryLoad: Cannot open file "C:\Users\adity\AppData\Roaming\VMware\c
2025-05-29T01:23:16.274Z In(05) vmx [msg.dictionary.load.openFailed] Cannot open file "C:\Users\adity\AppData
specified.
2025-05-29T01:23:16.274Z In(05) vmx PREF Optional preferences file not found at C:\Users\adity\AppData\Roaming
2025-05-29T01:23:16.285Z In(05) vmx lib/ssl: OpenSSL using default provider
```

Example log of Vmware Workstation

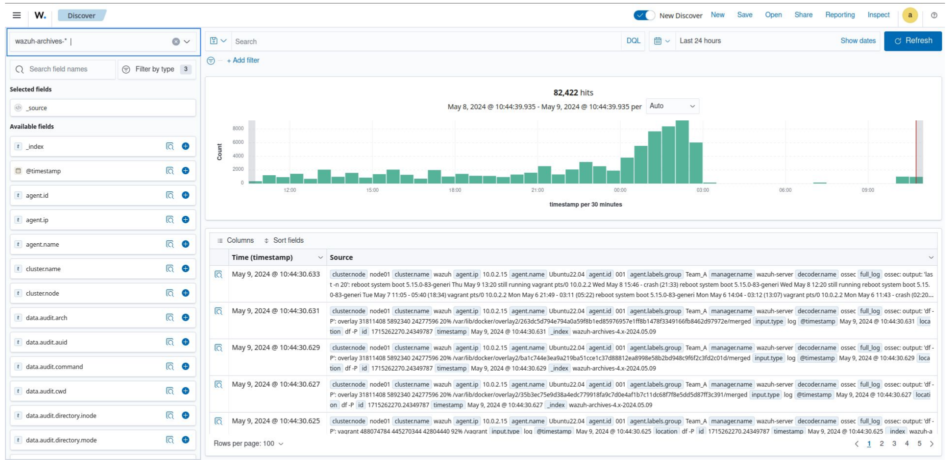
TAKE A BREAK !

- For your information, many application in industry is non standar, and don’t have logging function and make it harder doing DFIR

*My reaction for the situation

SIEM Role

- Central log collection, correlation
- Analytics for complex threat patterns
- Contextual alerts (Wazuh example)



Alert Wazuh from Data Source

Legacy Challenges

- No visibility or audit logs
- Data silos, uncoordinated tools
- Slow, reactive responses

*Situations after 3 months my team work together with this situation

TAKE A BREAK !

- Question : "Is it necessary for us to know what device is connected to the Switch Access on port 25, and is it necessary for us to understand the purpose of that device on that port?"

Modern SOC Benefits

- Unified visibility
- Fast response with workflows
- Team coordination, threat hunting

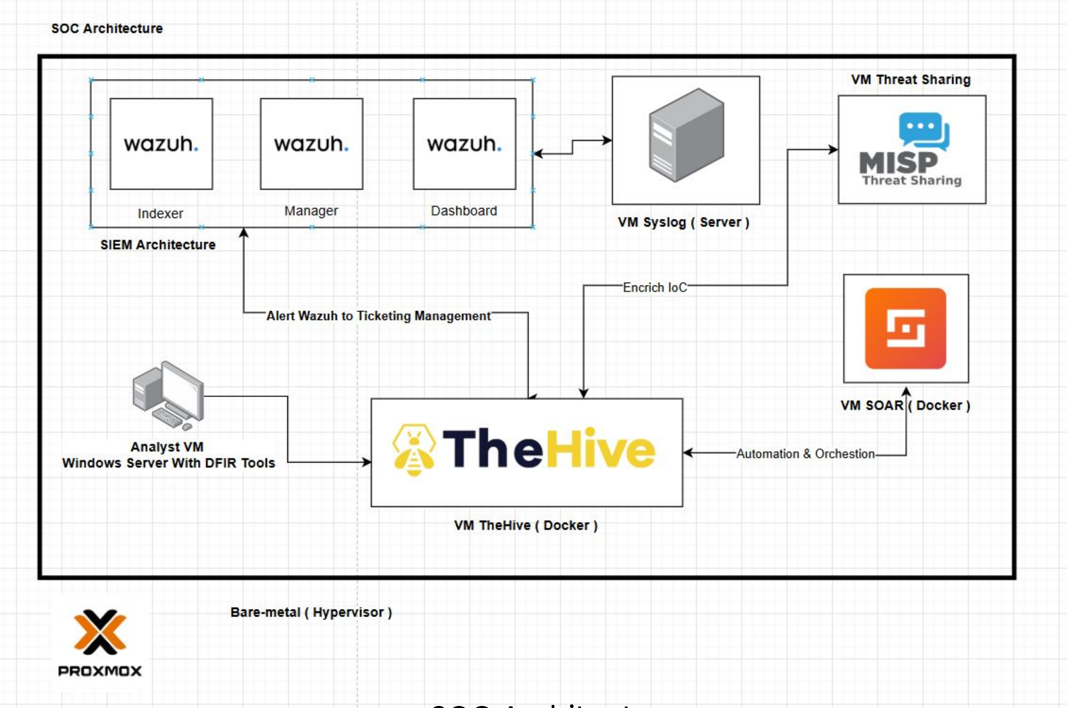
WHWOLOOWOWOO...... (Minimal ga mati 3 bulan sipp dah)

Guide To Make the labs

- "The document will be shared only in the webinar and is not intended to be shared with everyone."

Open-Source SOC Stack

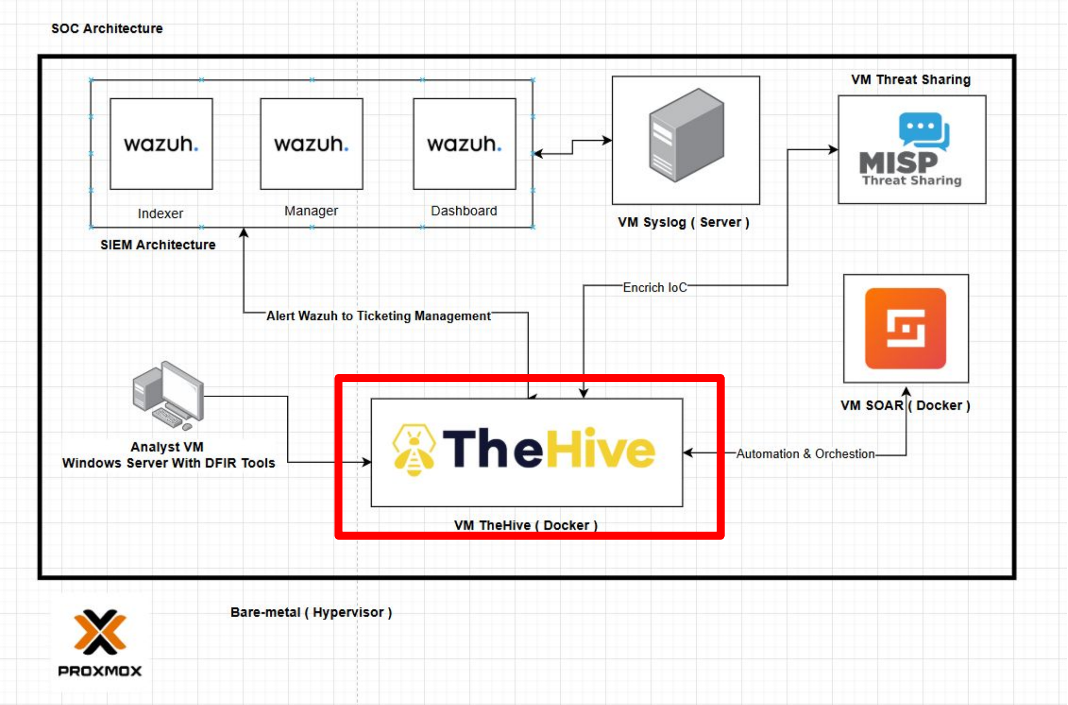
- SIEM: Wazuh
- Case: TheHive
- SOAR: Shuffle
- Intel: MISP
- Integrated & cost-effective



SOC Architecture

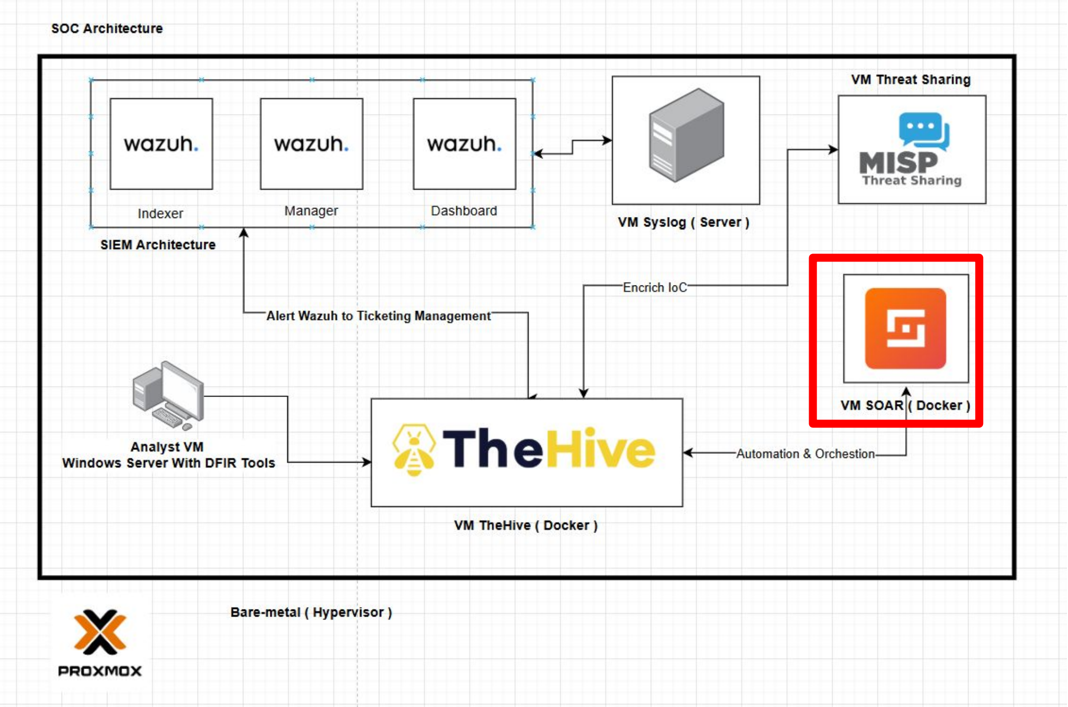
TheHive – Incident Management

- **Case management** with evidence, task tracking
- Collaboration among analysts
- Cortex & MISP integration
- Visibility & investigation timeline



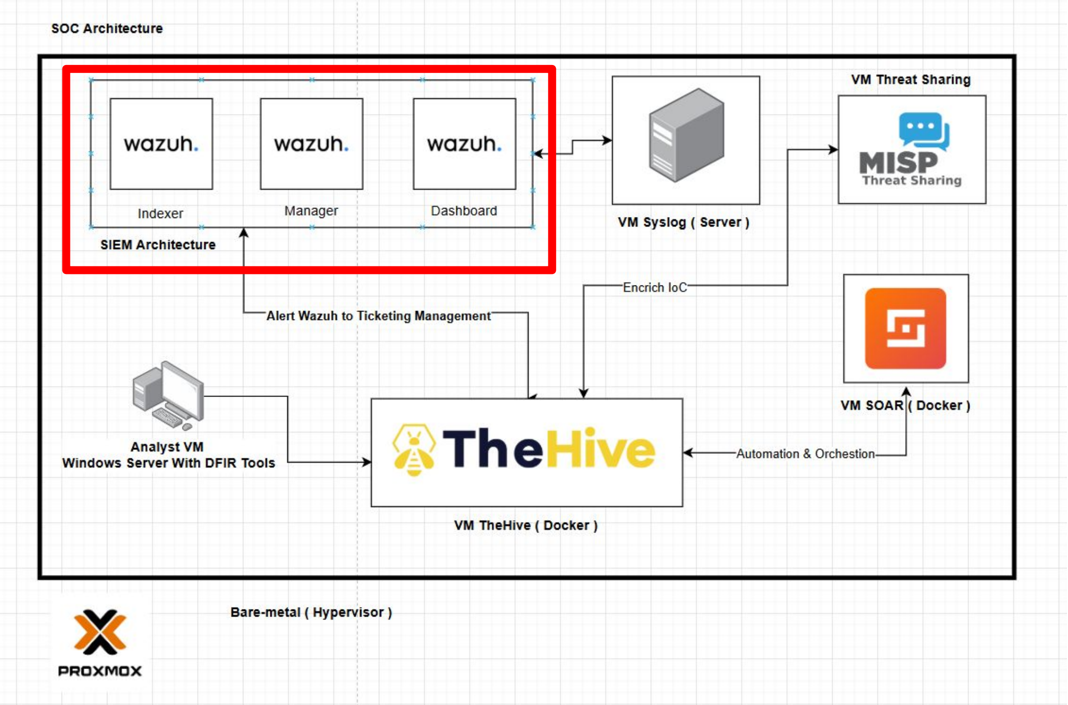
Shuffle – SOAR

- **Automates enrichment** & **response**
- Connects Wazuh → TheHive → MISP
- Email/Slack alerting
- Analyst efficiency booster



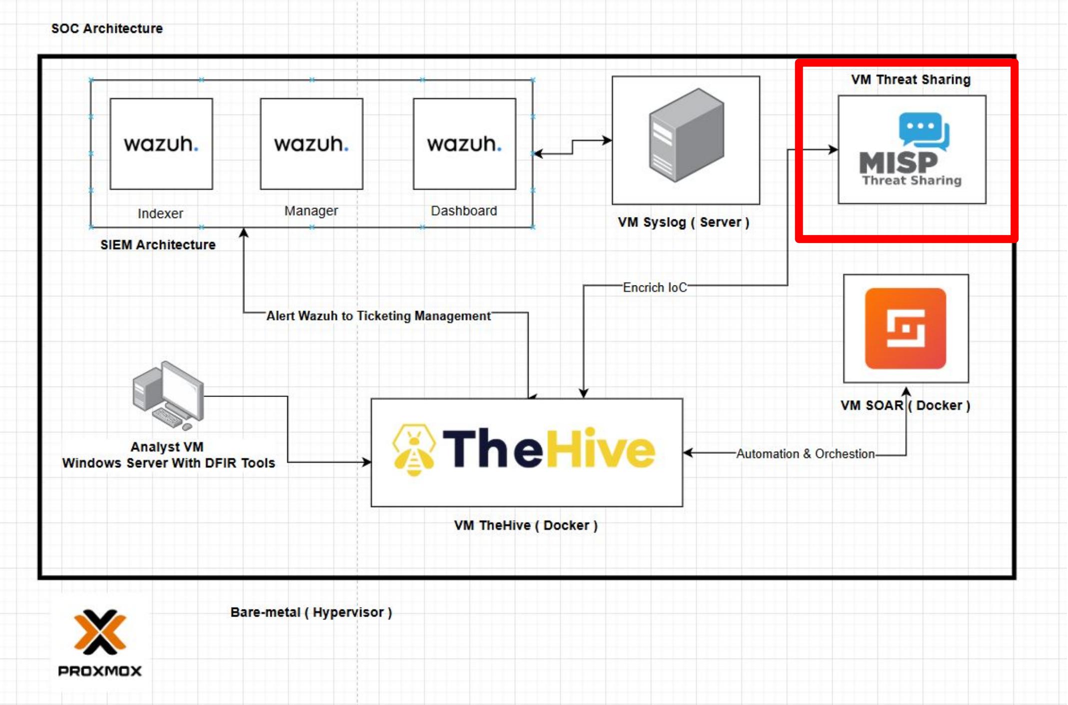
Wazuh – SIEM & HIDS

- **Collect logs from endpoint, network, cloud**
- Detect threats with custom **rules**
- Real-time alerting, dashboard
- API **integration** to SOAR



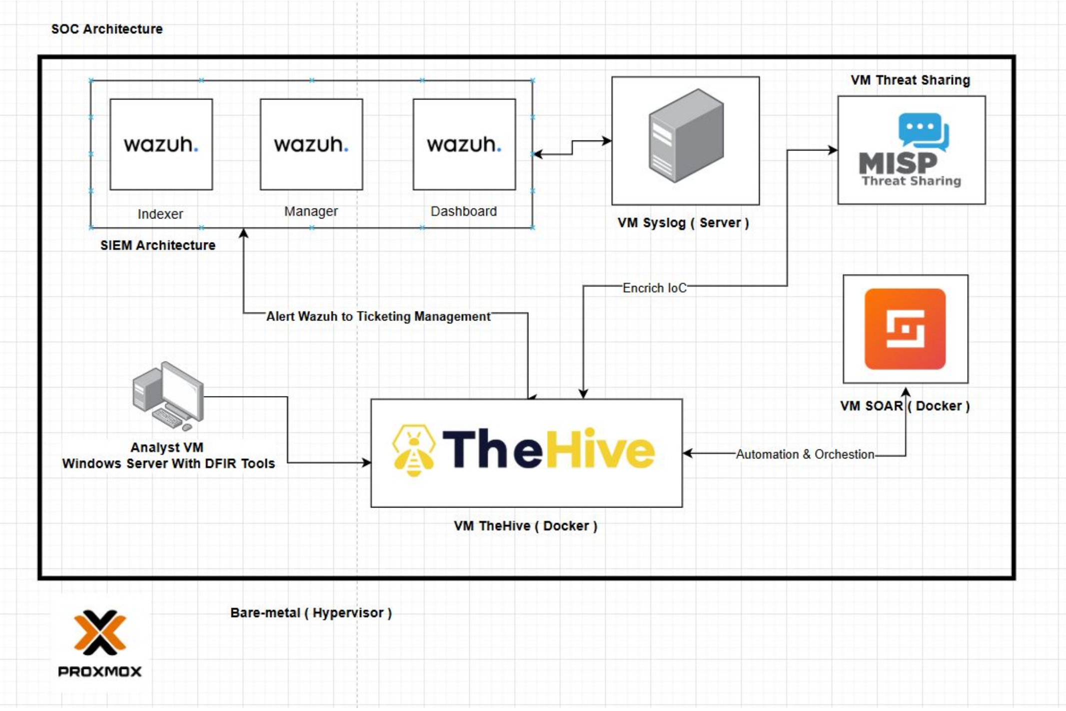
MISP – Threat Intelligence

- IOC repository: IP, hashes, domains
- Context: malware, MITRE ATT&CK
- Feeds Wazuh & TheHive
- Share intel with partners



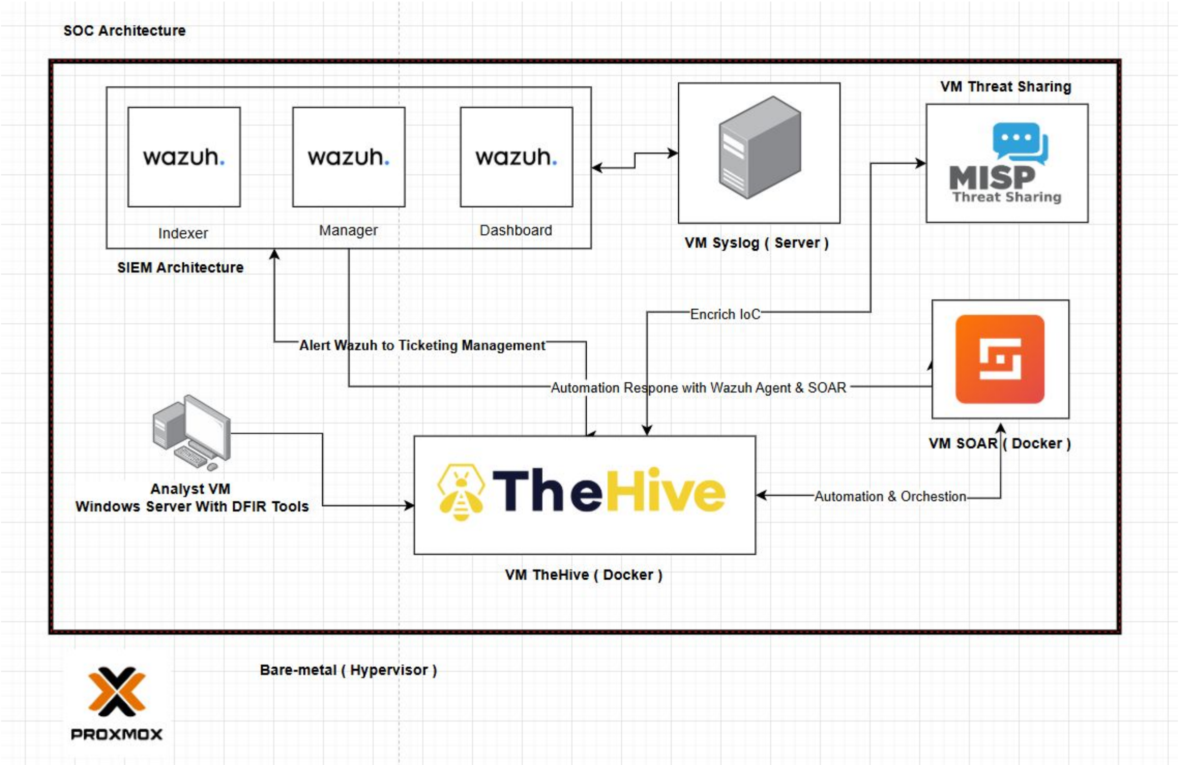
SOC Architecture & Data Flow

- Sources → Wazuh → Shuffle → TheHive
- MISP enriches at multiple points
- Analysts work in TheHive
- Feedback to MISP/rules



Workflow Example: Mimikatz

1. Alert from Wazuh
2. Trigger Shuffle
3. Enrich via VT/MISP
4. Case in TheHive
5. Notify SOC
6. Contain threat
7. Feed IOCs to MISP



SOC Metrics & Case Studies

- MTTD, MTTR, TP/FP rates
- IEEE: alert fatigue common
- SANS: automate junk triage
- Open-source SOCs proven effective

NIST IR Lifecycle

1. Preparation
2. Detection & Analysis
3. Containment & Recovery
4. Lessons Learned
5. Align playbooks to phases

SOC Playbooks

- Malware, phishing, ransomware
- Checklist per incident type
- Automate with Shuffle
- Update post-incident

Steps to Build SOC

1. Define scope & assets (DATA SOURCE)
2. Build team
3. Deploy open-source stack
4. Integrate workflows
5. Create playbooks
6. Go live & test
7. **Continuous improvement**

Key Takeaways

- People + Process + Tools
- Open-source = powerful SOC
- Centralized & automated
- Metrics drive maturity
- Begin small, scale smart

Reference

- SANS Guide to Security Operationsfile-5li8yy5lkirsxjnx9gdodcfile-5li8yy5lkirsxjnx9gdodc
- NIST SP 800-61r2 rapid7.com
- Check Point “What is a SOC?” checkpoint.com
- Dev.to Hughes Systique SOC article dev.todev.to
- Vielberth et al., “Security Operations Center: A Systematic Study…” (IEEE Access 2020) researchgate.netresearchgate.net
- JB Lemard-Reid, “Wazuh, TheHive, and Shuffle — SOC Automation Project” (Medium 2024) medium.commedium.com
- Atricore Open-Source SOC description atricore.com
- Darktrace blog on SOC Metrics (2024) darktrace.com; and other referenced materials throughout.

Q&A

- Ask about tools, scaling, workflows, metrics
- Discussion & support channels